

Group 1

Cervidae

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Introduction

The family *Cervidae* is a group of mammals commonly known as deer. These animals are known as even-toed ungulates with cloven hooves and slender legs. Another defining characteristic of this family is the presence of bony structures called antlers, which are grown and shed annually (Geist, 2026). As of 2026, there are about 60 species of deer distributed across Europe, Asia, North America, and South America, where they inhabit a wide range of habitats, ranging from tundra and forests to woodlands and tropical mountain forests (Alegsa, 2026). This family includes well-known species such as red deer (*Cervus elaphus*), elk (*Cervus canadensis*), white-tailed deer (*Odocoileus virginianus*), roe deer (*Capreolus capreolus*), and barking deer (*Muntiacus muntjak*). Mitochondrial cytochrome c oxidase subunit I (COI) is widely used in genetics because its mitochondrial sequence is shared across many groups yet different enough between species, making it a reliable marker for species identification and evolutionary studies. These species were selected as the focal taxa for examining evolutionary relationships by comparing their genetic sequences using FASTA. The giraffe (*Giraffa camelopardalis*), a member of the family *Giraffidae*, was chosen as the outgroup to serve as a reference point for the phylogenetic tree and for comparing the relationships between selected deer species.

Scientific name	Common name	NCBI accession	Length (bp)	Focal taxon / Outgroup
<i>Cervus elaphus</i>	Red Deer	KJ205551	618	Focal taxon
<i>Cervus canadensis</i>	Elk	MF632309	607	Focal taxon
<i>Odocoileus virginianus</i>	White-tailed Deer	KY033224	637	Focal taxon
<i>Capreolus capreolus</i>	Roe Deer	KY754490	655	Focal taxon
<i>Muntiacus muntjak</i>	Barking Deer	MT251413	660	Focal taxon
<i>Giraffa camelopardalis</i>	Giraffe	FJ958342	620	Outgroup

Methods

Mitochondrial cytochrome c oxidase subunit I (COI/COX1) sequences were retrieved from the NCBI Nucleotide database for five members of the family Cervidae—*Cervus elaphus*, *Cervus canadensis*, *Odocoileus virginianus*, *Capreolus capreolus*, and *Muntiacus muntjak*—along with *Giraffa camelopardalis* as an outgroup. These sequences were compiled into a multi-FASTA file and uploaded to the Galaxy platform (Usegalaxy.org). Multiple sequence alignment was performed using MAFFT with default parameters to ensure accurate positional homology across taxa. From the aligned dataset, a consensus sequence was generated to represent conserved regions. Phylogenetic inference was then conducted using FastTree, which applies approximately-maximum-likelihood methods to construct evolutionary trees from large alignments efficiently. The resulting tree was visualized within Galaxy, allowing interpretation of branching patterns and evolutionary relationships among the cervid species, with the giraffe serving as a reference point for divergence outside the family.

The following figures illustrate the key steps involved in constructing a phylogenetic tree, beginning with obtaining the FASTA file from NCBI and subsequently analyzing it using the Galaxy platform (Usegalaxy.org).

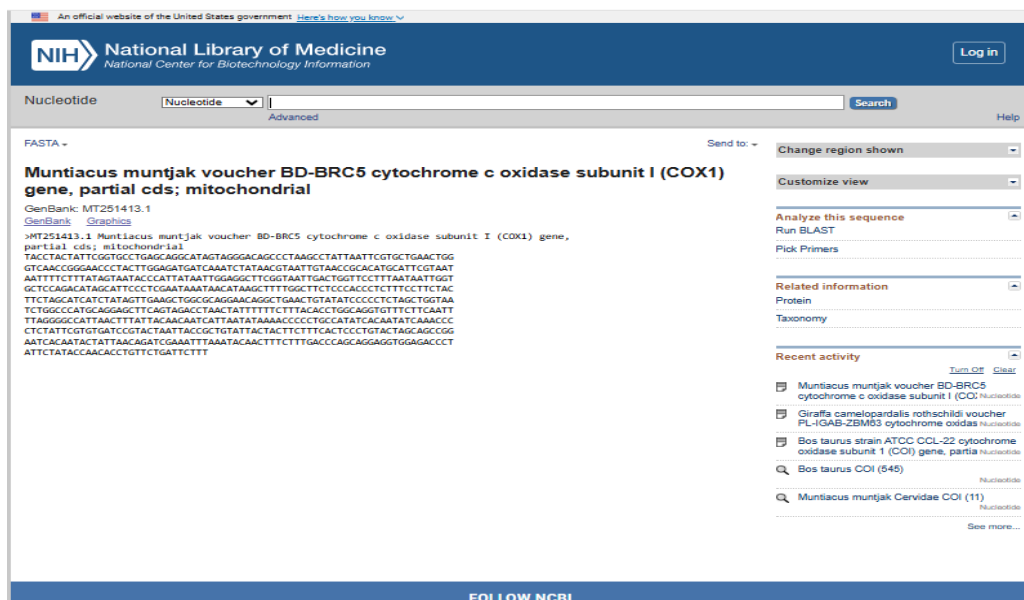


Figure 1. Download the COI gene sequence from the NCBI Nucleotide database in FASTA format, rename the file with your chosen organism's name, and use it as the input for the next steps in Galaxy.

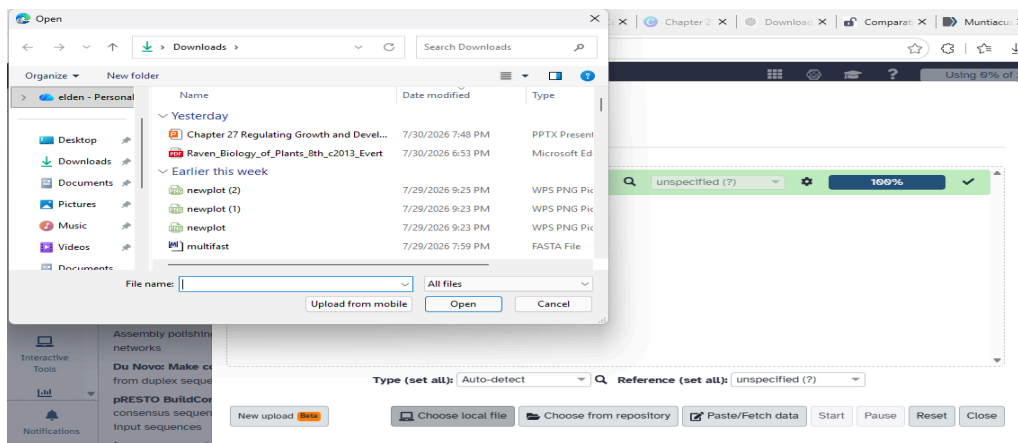


Figure 2. Upload the renamed FASTA file to Galaxy by selecting Upload, choosing Local File, and importing it for sequence alignment and phylogenetic tree construction.

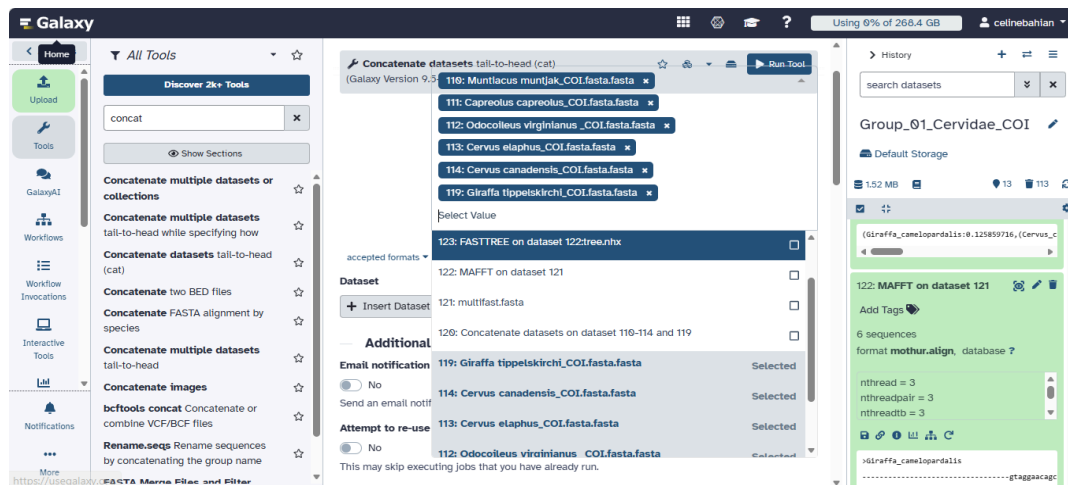


Figure 3. Use the Concatenate Sequences tool in Galaxy to combine your FASTA sequences into a multiFASTA file for multiple sequence alignment and phylogenetic tree construction.

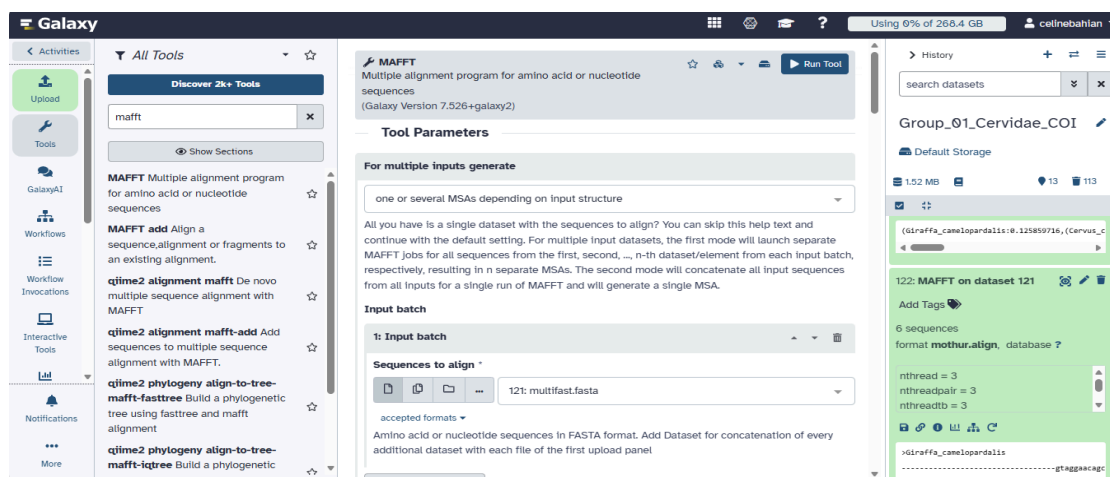


Figure 4. Run MAFFT in Galaxy, select the concatenated multiFASTA file as the input, and align the sequences to prepare them for phylogenetic analysis.

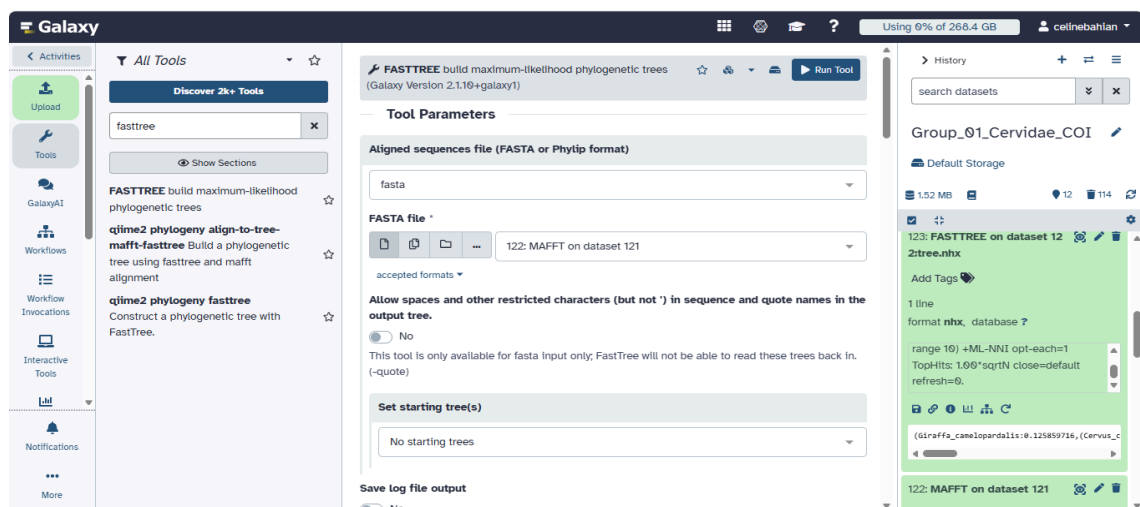


Figure 5. Use FastTree in Galaxy to generate a maximum-likelihood phylogenetic tree from the aligned sequences.

The screenshot displays the Galaxy web interface. On the left, a sidebar contains navigation links: Activities, Upload, Tools, GalaxyAI, Workflows, Workflow Invocations, Interactive Tools, Notifications, and More. The main workspace is titled '123: FASTTREE on dataset 122:tree.nhx'. It shows the tool's format as 'nhx', database as '?', and size as '249 b'. Below this, there are buttons for 'Preview', 'Visualize', 'Details', and 'Edit'. A search bar for visualizations is also present. The right sidebar shows a history of datasets, including 'Group_01_Cervidae_COI' and '123: FASTTREE on dataset 122:tree.nhx'.

Figure 6. Open the FastTree output in Galaxy to visualize the phylogenetic tree and examine the branching patterns and evolutionary relationships among the aligned sequences.

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>Giraffa_camelopardalis
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Consensus summary of conserved and variable sites

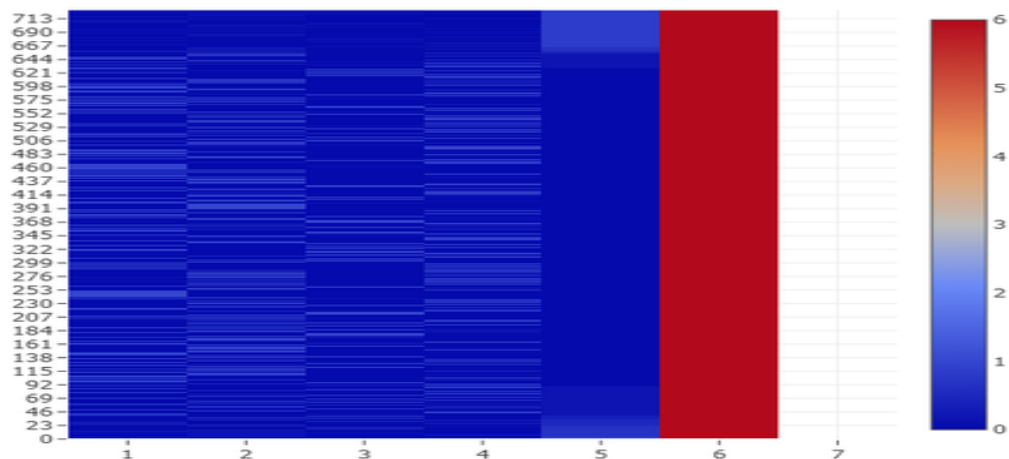


Figure 8. Heatmap of the MAFFT multiple sequence alignment showing nucleotide variation across the aligned sequences. Dark blue indicates conserved positions, while warmer colors represent regions with greater sequence variation.

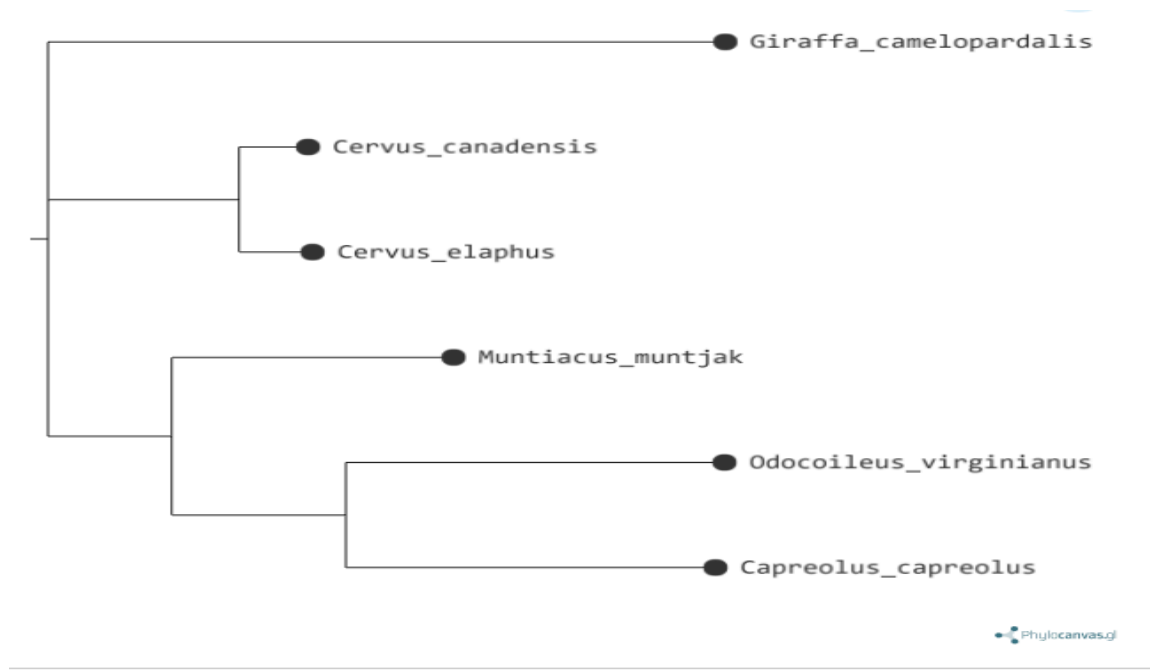


Figure 9. Phylogenetic tree of selected Cervidae species generated from mitochondrial COI sequences using the maximum-likelihood method. *Giraffa camelopardalis* was included as the outgroup.

The phylogenetic tree clearly separated the deer species (family Cervidae) from the outgroup, *Giraffa camelopardalis*. Among the deer, *Cervus canadensis* and *Cervus elaphus* clustered together, indicating that they are the most closely related species in this analysis and share a recent common ancestor. The short branches connecting these two species suggest that there is relatively little genetic difference between them based on the COI gene. In contrast, the longer branches leading to the other genera indicate greater genetic divergence. *Giraffa camelopardalis* was selected as the outgroup because it belongs to the family Giraffidae, which is closely related to Cervidae within the order Artiodactyla but is not part of the deer family.

Comparison with Published Studies

A study of Heckeberg, N. S. (2020) on the The systematics of the *Cervidae*: a total evidence approach, key phylogenetic findings include a confirmation of the two-subfamily classification: Cervinae (Muntiacini + Cervini) and Capreolinae (Alceini + Capreolini + Odocoileini + Rangiferini). All major clades (Cervinae, Cervini, Muntiacini, Capreolini, Odocoileini) recovered as monophyletic meaning that this group of organisms includes a common ancestor and all of its descendants. Heckeberg's results directly support the topology in the tree (revealed using FASTTREE). The two main clades (Cervinae and Capreolinae) also recovered with strong support. Within Cervinae, *Muntiacus* is sister to the *Cervus* clade, exactly as the tree shows. Within Capreolinae, *Capreolus* and *Odocoileus* are in separate tribes but grouped in the same subfamily. Heckeberg also specifically discusses Giraffa and Bovidae as outgroups within Ruminantia, which validate the choice of *Giraffa camelopardalis*.

On another study about Comparative genomics reveal phylogenetic relationship and chromosomal evolutionary events of eight Cervidae species, published by Tang, L., Dong, S., & Xing, X. in 2024, it uses chromosome-level whole-genome assemblies from eight Cervidae species (plus cattle as an outgroup) to resolve phylogeny and trace chromosomal evolution. The study finds phylogenetic tree built from 2,480 single-copy orthologous genes confirms two main branches: Capreolinae (reindeer) and Cervinae (Muntiacini + Cervini). And within Cervini, *Cervus canadensis* (wapiti) and *Cervus elaphus* (red deer) are sister taxa. Gene family evolution finds significant contraction of the olfactory receptor gene family across all eight Cervidae species, possibly related to habitat fragmentation, and significant expansion of the histone gene family which may have been involved in chromatin stability and gene regulation. Tang et al. (2024) provides whole-genome-level confirmation of the same two-subfamily topology the tree shows (recovered from FASTTREE). Their placement of *Cervus canadensis* and *Cervus elaphus* as sister species within Cervini matches the tree exactly. It also confirms *Muntiacus* as sister to Cervini within Cervinae.

Heckeberg included more deer species and fossil data, while Tang et al. used genome-wide data to better understand deer evolution. Both studies support the relationships shown in the phylogenetic tree.

Discussion of Agreements, Disagreements, and Limitations of Using COI alone

The findings of this study are generally consistent with previous molecular phylogenetic studies. In particular, *Cervus canadensis* and *Cervus elaphus* were grouped together, supporting their close evolutionary relationship. *Giraffa camelopardalis* also served as an appropriate outgroup because it belongs to the family Giraffidae, which is outside Cervidae. However, the positions of *Muntiacus muntjak*, *Odocoileus virginianus*, and *Capreolus capreolus* differed slightly from those reported published phylogenies by Heckeberg, N. S. (2020). These differences may be due to the use of only the mitochondrial COI gene in this analysis. Although COI is widely used for species identification and phylogenetic studies, relying on a single maternally inherited gene provides limited information. Including additional mitochondrial and nuclear genes would likely improve the resolution and accuracy of the inferred evolutionary relationships.

References

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- Tang, L., et al. (2024). Comparative genomics reveal phylogenetic relationship and chromosomal evolutionary events of eight Cervidae species. Animals, 14(7), 1063. <https://www.mdpi.com/2076-2615/14/7/1063>